Team 19: Does the Price of Gold Kill Brides?

1. Main Hypothesis and Research Questions

When gold becomes more expensive, do more Indian brides die over dowry? Gold jewellery is central to many Indian dowry practices. As gold prices rise, meeting a groom's family's expectations may become harder for the bride's family, potentially increasing harassment and, in extreme cases, deaths within marriage.

The raw national trends appear to contradict this hypothesis. Gold in Mumbai rose from approximately Rs. 4,579 per 10g in 2001–02 to Rs. 75,842 in 2024–25 (RBI). Recorded dowry deaths increased from approximately 6,851 in 2001 to a peak of 8,618 in 2011, then fell to 5,737 in 2024 (NCRB). A naive correlation might suggest that costlier gold saves brides. Our project investigates whether this pattern is causal, coincidental, or driven by other trends.

We investigate three questions:

1. **Shocks, not trends:** After adjusting for inflation and removing long-run trends, are sharp gold-price increases followed by more dowry deaths? Since Section 304B concerns deaths within seven years of marriage, we will test lagged and multi-year-average gold prices.
2. **Which states are the main culprits?** Do states with historically high dowry-death rates respond more strongly to gold-price changes than other states?
3. **Are gold-price changes associated with dowry-death rates 1–2 years later?** Is there any difference in our interpretations using data lagged by 2 years or so? We will perform the same analysis but on data lagged by 2 years to see if the association is real.

2. Sources and Related Work

- **Core paper:** Bhalotra, Chakravarty and Gulesci (2020), "The Price of Gold: Dowry and Death in India", *Journal of Development Economics*.
- **Crime data:** NCRB, *Crime in India* reports (2001–2024), chapters on crimes against women.

3. Datasets

Dataset	Variables	Years
Kaggle: Crimes Against Women in India	Dowry deaths; Dowry Prohibition Act cases by state	2001–2021
NCRB 2022, 2023, and 2024	State dowry deaths; female population for rates	2020–2024
RBI Handbook, Table 35	Mumbai gold price, Rs. per 10g	1985–86 to 2025–26
Kaggle: Gold Price India	Daily and annual gold prices in INR	1978–Jan. 2021
RBI Handbook, Table 9 and Table 34	State per-capita income; CPI	2001–2026

4. Methods, Tools, and Visualisation

We will use Python (`pandas`, `statsmodels`, `scipy.stats`, `matplotlib`, and Plotly) for data cleaning, merging, detrending, lagged regressions, and state fixed-effects models.

5. LLM and Agent Use

We used ChatGPT to brainstorm topics, run agents that checked dataset availability for this and two discarded topics, locate the Bhalotra et al. paper, and draft this proposal.

Transcript: https://chatgpt.com/share/6accda5f-644c-83ee-85e2-4fb48f124d57

6. Logistics and Work Plan

We will add our team number, title, GitHub link, and three primary sources (NCRB, RBI, Kaggle) to the course response spreadsheet. The project will use a shared GitHub repository with a shared log of data-cleaning decisions. We will meet online once a week to check on our progress.

Work	Assigned to	Tentative Deadline
Gold Price Data Collection & Analysis		
Collect historical gold prices from RBI/World Bank, clean the data, calculate annual percentage changes, and plot gold-price trends.	Members 1 and 2	18 Oct.
Dowry Death Data Collection & Analysis		
Collect NCRB dowry-death statistics, organise state-wise and year-wise data, calculate dowry-death rates where population data are available, and visualise trends.	Members 3 and 4	18 Oct.
Checkpoint 1		
GitHub repository and exploration code.	All	21 Oct.
Statistical Analysis		
Combine both datasets, align the years, perform Pearson correlation and linear regression, investigate 1–2 year lagged correlations, and compare results across states where feasible.	Members 1 and 2	25 Oct.
Literature Review & Interpretation		
Study relevant research papers, interpret the statistical results, investigate possible explanations for observed relationships, discuss limitations, and coordinate the final report and presentation.	Members 3 and 4	29 Oct.
Deliverables		
Prepare and finalise the report, 3–5 minute video, and any other required deliverables.	All	2 Nov.
Checkpoint 2		
Submit the final report, video, and references.	All	4 Nov.

Logistics: Tentative Work Schedule (click here!)